How Numbers Work

Sets, Properties and Extensions

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Legend

Theory

Remarks

Problems

A structured exploration of number systems for all the dedicated self-learners

Preface

Introduction

Numbers form a foundational layer of mathematical thought. This handout presents number systems as a sequence of necessary extensions, each introduced to resolve a limitation of the previous one.

Sets

A **set** is a collection of objects, called *elements*. If x belongs to a set A, we write:

$$x \in A$$

A Note on Relations

A **relation** on a set assigns a condition between elements. A special type, called an **equivalence relation**, satisfies:

For $a, b, c \in \mathbb{N}$, equality defines an equivalence relation:

- $a = a$ (reflexivity)
- $a = b \implies b = a$ (symmetry)
- $a = b,\ b = c \implies a = c$ (transitivity)

Such relations group elements into **equivalence classes**, where elements are considered "the same" under the relation.

Remarks

We will use equivalence classes of pairs $(a, b) \in \mathbb{N} \times \mathbb{N}$ to construct a system where subtraction is always possible.

The Culprits

We will work with five primary number systems:

- $\mathbb{N}$ — Natural Numbers
- $\mathbb{Z}$ — Integers
- $\mathbb{Q}$ — Rational Numbers
- $\mathbb{R}$ — Real Numbers
- $\mathbb{C}$ — Complex Numbers

Remarks

Each system arises to fix a limitation:

$$\mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R} \to \mathbb{C}$$

Meet the Naturals

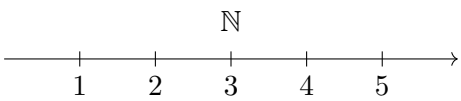
Theory

The natural numbers are the most intuitive number system:

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

They satisfy:

- **Unboundedness:** For any $M \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $n > M$.
- **Well-ordering:** Every non-empty subset of $\mathbb{N}$ has a least element.
- **Closure:** If $a, b \in \mathbb{N}$, then $a + b \in \mathbb{N}$ and $ab \in \mathbb{N}$.
- **Commutativity:**
$$a + b = b + a, \quad ab = ba$$
- **Associativity:**
$$a + (b + c) = (a + b) + c$$
- **Distributivity:**
$$a(b + c) = ab + ac$$



Remarks

The natural numbers support addition and multiplication, but not subtraction in general. This limitation motivates the construction of the integers $\mathbb{Z}$.

Problems

Where $\mathbb{N}$ Breaks

1. For which natural numbers $a, b \in \mathbb{N}$ is $a - b$ also in $\mathbb{N}$? What happens when $b > a$?
2. Consider the equation:
$$x + 3 = 1$$
Does this have a solution in $\mathbb{N}$?
3. Try to solve:
$$x + a = b$$

for arbitrary $a, b \in \mathbb{N}$. When does a solution exist in $\mathbb{N}$?

4. Define a relation on pairs $(a, b) \in \mathbb{N} \times \mathbb{N}$:

$$(a, b) \sim (c, d) \iff a + d = b + c$$

   - Show this is an equivalence relation.
   - Compute:

$$(3, 1), \quad (4, 2), \quad (5, 3)$$

     What do you notice?

5. Consider pairs (a, b) as representing a "difference" $a - b$. Show that:

$$(a, b) \sim (c, d) \implies a - b = c - d$$

6. Define addition on pairs:

$$(a, b) + (c, d) = (a + c, b + d)$$

   Show that this respects the relation $\sim$.

7. Find a pair (a, b) such that:

$$(a, b) + (b, a) \sim (0, 0)$$

   What does this suggest?

8. Is there a pair (a, b) such that:

$$(a, b) \sim (0, 0)$$

   What does this represent?

9. Show that for any (a, b), there exists another pair that "cancels" it under addition.
10. Challenge: Construct a system of numbers using pairs $(a, b) \in \mathbb{N} \times \mathbb{N}$ that allows subtraction to always be possible. Define it like we defined N.

Meet the Integers

Theory

The natural numbers fail to solve equations of the form:

$$a + x = b \quad \text{when } b < a$$

To resolve this, we extend $\mathbb{N}$ to a larger system:

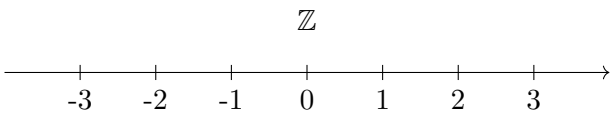
$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

What $\mathbb{Z}$ fixes and introduces:

1. **Solvability of Equations**
   For all $a, b \in \mathbb{Z}$, the equation
   $$a + x = b$$
   always has a solution:
   $$x = b - a$$
2. **Bidirectional Infinity**
   Unlike $\mathbb{N}$, which extends only to the right, $\mathbb{Z}$ extends infinitely in both directions:
   $$\cdots < -2 < -1 < 0 < 1 < 2 < \ldots$$
3. **Order but not Well-Ordering**
   $\mathbb{Z}$ is ordered, but not well-ordered: there is no smallest element due to negative infinity.
4. **Successors and Predecessors**
   Every integer has both:
   $$\text{successor: } n + 1, \quad \text{predecessor: } n - 1$$
5. **Closure under Addition and Subtraction**
   For all $a, b \in \mathbb{Z}$:
   $$a + b \in \mathbb{Z}, \quad a - b \in \mathbb{Z}$$
   In particular, subtraction is now always possible.
6. **Sign Structure**
   For $a, c \in \mathbb{N}$, define $x = -a$. Then:
   $$x + c = (-a) + c = \begin{cases} c - a & \text{if } c \geq a \\ -(a - c) & \text{if } a > c \end{cases}$$
7. **Algebraic Laws Still Hold**
   For all $a, b, c \in \mathbb{Z}$:
   Commutativity:
   $$a + b = b + a, \quad ab = ba$$
   Associativity:
   $$a + (b + c) = (a + b) + c$$
   Distributivity:
   $$a(b + c) = ab + ac$$
8. **Identity Elements**
   There exist special elements:
   $$a + 0 = a, \quad a \cdot 1 = a$$
9. **Additive Inverses**

For every $a \in \mathbb{Z}$, there exists $-a \in \mathbb{Z}$ such that:

$$a + (-a) = 0$$



Remarks

The integers do more than fix subtraction in $\mathbb{N}$.
They ensure that equations of the form $a + x = b$ are always solvable, even when both a and b lie outside $\mathbb{N}$.
However, this extension comes with a tradeoff: $\mathbb{Z}$ is no longer well-ordered.
This reflects a general principle:
Extending a number system increases solvability, but may sacrifice certain structural properties.

Problems

What $\mathbb{Z}$ Fixes

1. Consider the equation:
$$x + 5 = 2$$
Explain why this has no solution in $\mathbb{N}$, but does in $\mathbb{Z}$.
2. More generally, for $a, b \in \mathbb{N}$, determine when:
$$x + a = b$$
has a solution in $\mathbb{N}$. What must be added to always solve it?
3. Show that for every $a \in \mathbb{Z}$, there exists $x \in \mathbb{Z}$ such that:
$$a + x = 0$$
What does this element represent?
4. Prove that subtraction can be rewritten as:
$$a - b = a + (-b)$$
Why is this not always possible in $\mathbb{N}$?
5. Show that for any $a, b \in \mathbb{Z}$:
$$a - b = 0 \iff a = b$$
6. Prove that:
$$-(a + b) = (-a) + (-b)$$
7. Show that:
$$(-1) \cdot a = -a$$

for all $a \in \mathbb{Z}$.

8. Prove that:
$$(-a)(-b) = ab$$
9. Show that if $a + b = a + c$, then $b = c$. (Why does this rely on additive inverses?)
10. Prove that $\mathbb{Z}$ is closed under subtraction. Why is this property new compared to $\mathbb{N}$?
11. Show that $\mathbb{Z}$ is not well-ordered. (Find a subset with no least element.)
12. Show that for every $n \in \mathbb{Z}$, both $n + 1$ and $n - 1$ exist in $\mathbb{Z}$. Why is this different from $\mathbb{N}$?
13. Prove that:
$$|a + b| \leq |a| + |b|$$
for all $a, b \in \mathbb{Z}$.
14. Show that:
$$a^2 \geq 0$$
for all $a \in \mathbb{Z}$, and determine when equality holds.
15. Challenge: Show that there is no integer between n and $n + 1$ for any $n \in \mathbb{Z}$. Compare this with what happens in $\mathbb{Q}$.

Remarks

The rationals do more than fix division in $\mathbb{Z}$.
They ensure that equations of the form $ax = b$ are always solvable for $a \neq 0$.
Unlike $\mathbb{Z}$, this extension does not sacrifice order, but introduces a new phenomenon: between any two rational numbers, there are infinitely many others.
This reflects a general pattern:
Each extension increases solvability and density, while preserving as much structure as possible.

Meet the Rationals

Theory

The integers solve subtraction, but still fail to solve equations of the form:

$$ax = b \quad \text{when } a \nmid b$$

To resolve this, we extend $\mathbb{Z}$ to:

$$\mathbb{Q} = \left\{ \frac{a}{b} \;\middle|\; a, b \in \mathbb{Z},\ b \neq 0 \right\}$$

Definition and Structure

1. **Fractions as Solutions**

The expression $\frac{a}{b}$ is defined to be the solution of:

$$bx = a$$

Thus, division is introduced as the inverse of multiplication.

2. **Equivalence of Fractions**
   Two fractions represent the same number if:

$$\frac{a}{b} = \frac{c}{d} \iff ad = bc$$

   This ensures consistency of representation.

3. **Scaling Property**
   For any nonzero $p \in \mathbb{Z}$:

$$\frac{a}{b} = \frac{pa}{pb}$$

   Thus, a rational number has infinitely many representations.

4. **Sign Convention**

$$\frac{-a}{b} = \frac{a}{-b}$$

   We usually write the negative sign in the numerator:

$$-\frac{a}{b}$$

   If both are negative:

$$\frac{-a}{-b} = \frac{a}{b}$$

5. **Operations are Well-Defined**
   Addition:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

   Multiplication:

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

   These are **well-defined**, meaning: they do not depend on the chosen representation.

6. **Why Multiplication is Meaningful**
   Since $\frac{a}{b}$ represents the solution to $bx = a$, multiplying two rationals corresponds to combining two such scalings:

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

   This preserves the structure of solving equations of the form $ax = b$.

7. **Closure**
   For all $x, y \in \mathbb{Q}$:

$$x + y \in \mathbb{Q}, \quad xy \in \mathbb{Q}, \quad x \neq 0 \implies \frac{1}{x} \in \mathbb{Q}$$

8. **Algebraic Laws**
   All familiar laws extend from $\mathbb{Z}$:

- Commutativity
- Associativity
- Distributivity

9. **Identity Elements**

$$x + 0 = x, \quad x \cdot 1 = x$$

10. **Multiplicative Inverses**
For $x \neq 0$:

$$x \cdot \frac{1}{x} = 1$$

Remarks

The rationals introduce a major structural shift:

- **Density:** Between any two rational numbers, there are infinitely many others. Unlike $\mathbb{N}$ and $\mathbb{Z}$, $\mathbb{Q}$ has no notion of a "next" element.
- **Arbitrary Precision:** Rational numbers can approximate quantities as closely as desired:
$$\frac{1}{3}, \frac{33}{100}, \frac{333}{1000}, \ldots$$
- **Closure of Division:** For $a \neq 0$, the equation $ax = b$ always has a solution in $\mathbb{Q}$.
- **Multiple Representations:** Each rational number has infinitely many forms:
$$\frac{1}{2} = \frac{2}{4} = \frac{50}{100}$$

We also, for rigour and the topic's interestingness seemingly, include:

Theory

Density of $\mathbb{Q}$
Let $x, y \in \mathbb{Q}$ with $x < y$. Then there exists $r \in \mathbb{Q}$ such that:

$$x < r < y$$

Proof:
Take:

$$r = \frac{x + y}{2}$$

Since $\mathbb{Q}$ is closed under addition and division by 2:

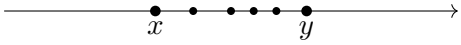
$$r \in \mathbb{Q}$$

Also:

$$x < \frac{x + y}{2} < y$$

Thus, there exists at least one rational between any two rationals.
Repeating this process shows there are infinitely many.

Infinitely many points



Problems

Why $\mathbb{Q}$ Exists

1. Consider the equation:
$$2x = 1$$
Does this have a solution in $\mathbb{Z}$? What kind of number would solve it?
2. More generally, for $a, b \in \mathbb{Z}$, when does:
$$ax = b$$
have a solution in $\mathbb{Z}$?
3. Show that if $a \nmid b$, then no integer solution exists to $ax = b$. What does this suggest we need to add?
4. Consider two pairs (a, b) and (c, d) with $b, d \neq 0$. Define:
$$(a, b) \sim (c, d) \iff ad = bc$$
   - Show this is an equivalence relation.
   - What does each equivalence class represent?
5. Show that:
$$\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$$
Explain why infinitely many representations exist.
6. Show that:
$$\frac{a}{b} = \frac{pa}{pb}$$
for any nonzero $p \in \mathbb{Z}$.
7. Show that:
$$\frac{-a}{b} = \frac{a}{-b}$$
and explain why we usually keep the negative sign in the numerator.
8. Define addition on pairs:
$$(a, b) + (c, d) = (ad + bc, bd)$$
Show that this respects the relation $\sim$.
9. Define multiplication on pairs:
$$(a, b)(c, d) = (ac, bd)$$
Show that this is well-defined.
10. Find a rational number strictly between:
$$\frac{1}{2} \quad \text{and} \quad \frac{2}{3}$$

Can you find infinitely many?

11. Show that between any two rational numbers $x < y$, there exists another rational number.
12. Challenge: Find a rational number between x and y that is **not** their midpoint.
13. Show that no matter how close two rational numbers are, there is always another rational between them.
14. Show that $\mathbb{Q}$ has no "next" element: there is no smallest rational greater than a given rational.
15. Challenge (important): Is there a rational number whose square is 2? What does your answer suggest?

Remarks

Despite their richness, the rationals are still incomplete.
There exist numbers that cannot be expressed as $\frac{a}{b}$:

$$x^2 = 2 \quad \text{has no solution in } \mathbb{Q}$$

Thus, even though $\mathbb{Q}$ is dense, it still contains "gaps".
This motivates the next extension: the real numbers $\mathbb{R}$.

Meet the Reals

A Fundamental Limitation of $\mathbb{Q}$

Claim: $\sqrt{p}$ is irrational for any prime p.

Proof:

Assume $\sqrt{p} = \frac{a}{b}$ in lowest terms, with $a, b \in \mathbb{Z}$.

Then:

$$p = \frac{a^2}{b^2} \implies a^2 = pb^2$$

So $p \mid a^2 \implies p \mid a$.

Let $a = pk$. Then:

$$a^2 = p^2k^2 = pb^2 \implies pk^2 = b^2$$

Thus $p \mid b^2 \implies p \mid b$.

So p divides both a and b, contradicting that $\frac{a}{b}$ was in lowest terms.

Hence $\sqrt{p} \notin \mathbb{Q}$.

Irrational Numbers

A number is called **irrational** if it cannot be written as:

$$\frac{a}{b}, \quad a, b \in \mathbb{Z}$$

Thus:

$$\mathbb{R} = \mathbb{Q} \cup \{\text{irrational numbers}\}$$

The Number Line

The real numbers correspond to points on a continuous line.

Unlike $\mathbb{Q}$, which leaves gaps, $\mathbb{R}$ fills every point.

Thus, $\mathbb{R}$ represents the **maximal extension of number as magnitude**.

Dedekind's Construction

Early Mathematician and very interesting and influential person Richard Dedekind defined Real Nos. like this.

A real number can be defined as a partition of $\mathbb{Q}$ into two sets:

$$\mathbb{Q} = L \cup U$$

such that:

- L contains all numbers less than the real number
- U contains all numbers greater than it
- L has no greatest element

Example: $\sqrt{2}$ corresponds to:

$$L = \{x \in \mathbb{Q} : x^2 < 2\}, \quad U = \{x \in \mathbb{Q} : x^2 > 2\}$$

Theory

The real numbers extend $\mathbb{Q}$ by eliminating all gaps.

$$\mathbb{R} = (-\infty, \infty)$$

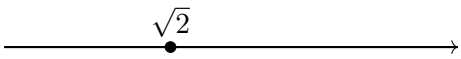
What $\mathbb{R}$ fixes and introduces:

1. **Completeness**
   Every bounded set has a least upper bound.
2. **No Gaps**
   All limits of converging sequences in $\mathbb{Q}$ exist in $\mathbb{R}$.
3. **Closure**

For $x, y \in \mathbb{R}$:

$$x + y,\ xy \in \mathbb{R}$$

4. **Order**
   $\mathbb{R}$ is totally ordered.
5. **Algebraic Laws**
   All standard laws hold.
6. **Continuity**
   $\mathbb{R}$ forms a continuous number line.



No gaps in $\mathbb{R}$

Remarks

The real numbers represent a fundamental leap.
While $\mathbb{Q}$ is dense, it is still incomplete. $\mathbb{R}$ removes this limitation entirely.
Uncountability:
The set $\mathbb{Q}$ is countable, but $\mathbb{R}$ is not.
This means there are strictly more real numbers than rational numbers.
Intuitively: no matter how many rationals we list, there will always be real numbers missing.
Thus, most real numbers are irrational.
Significance:
$\mathbb{R}$ is the foundation of:

- calculus
- analysis
- geometry

For a small taste of higher mathematics we include:

Remarks

Idea of Uncountability (Diagonal Argument)
Assume all real numbers in $(0, 1)$ can be listed.
Construct a new number by changing the n-th digit of the n-th number.
This new number differs from every number in the list.
Thus, no list can capture all real numbers.

Problems

Why $\mathbb{R}$ Exists

1. Show that there is no rational number whose square is 2. What does this imply about $\mathbb{Q}$?
2. Consider the set:
$$S = \{x \in \mathbb{Q} : x^2 < 2\}$$

- Show that S is bounded above in $\mathbb{Q}$.
- Does S have a least upper bound in $\mathbb{Q}$?

What does this suggest?

3. Find a sequence of rational numbers that gets closer and closer to $\sqrt{2}$. Does it ever reach it in $\mathbb{Q}$?
4. Show that between any two real numbers, there exists a rational number. (Why does this not contradict density of $\mathbb{Q}$?)
5. Show that between any two real numbers, there exists an irrational number.
6. Show that $\mathbb{R}$ has no "gaps": intuitively explain why every point on the number line corresponds to a real number.
7. Consider a bounded set $A \subset \mathbb{R}$. What might it mean for A to have a "least upper bound"?
   Give examples.
8. Find a set of rational numbers that is bounded but has no maximum.
   What happens to this set in $\mathbb{R}$?
9. Show that:
$$|x| = 0 \iff x = 0$$
   for all $x \in \mathbb{R}$.
10. Show that:
$$|x + y| \leq |x| + |y|$$
   (Why is this inequality important?)
11. Show that if $x^2 < 0$, then $x \notin \mathbb{R}$. What does this suggest about the next extension?
12. Challenge: Find two real numbers $x < y$ such that there are infinitely many rational numbers and infinitely many irrational numbers between them.
13. Challenge (deep): Construct a number by taking an infinite decimal:
$$0.101001000100001\ldots$$
   Is this rational or irrational?

Meet the Complexes

The real numbers $\mathbb{R}$ resolve division and eliminate gaps, but still fail at a fundamental level:

$$x^2 + 1 = 0$$

There is no real number whose square is negative.

Thus $\mathbb{R}$ is not sufficient to solve all polynomial equations.

We now ask:

Can we extend $\mathbb{R}$ in the smallest possible way so that this equation has a solution?

Theory

Step 1: Forcing a Solution
Introduce a new element i such that:

$$i^2 = -1$$

This is the minimal extension required to solve:

$$x^2 + 1 = 0$$

Step 2: What Should Numbers Look Like Now?
If i exists, then expressions like $a + bi$ must be included.
To remain closed under addition and multiplication, we consider all such expressions:

$$z = x + iy, \quad x, y \in \mathbb{R}$$

Thus:

$$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\}$$

Key Insight
A complex number is an ordered pair (x, y) written as $x + iy$.
Step 3: Defining Addition
To extend real addition consistently:

$$(a, b) + (c, d) = (a + c, b + d)$$

Thus:

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

Step 4: Defining Multiplication
We require:

- distributivity
- $i^2 = -1$

Compute:

$$(a + bi)(c + di) = ac + adi + bci + bdi^2$$

Using $i^2 = -1$:

$$= (ac - bd) + (ad + bc)i$$

This is the only rule that preserves the required properties.
Step 5: Algebraic Structure
For all $z_1, z_2, z_3 \in \mathbb{C}$:

- Commutativity:
$$z_1 + z_2 = z_2 + z_1, \quad z_1 z_2 = z_2 z_1$$
- Associativity:
$$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$$

- Distributivity:

$$z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$$

Step 6: Embedding $\mathbb{R}$
Every real number appears inside $\mathbb{C}$:

$$x = x + 0i$$

Thus:

$$\mathbb{R} \subset \mathbb{C}$$

Step 7: Identity Elements

$$z + 0 = z, \quad z \cdot 1 = z$$

Real and Imaginary Parts

Let:

$$z = x + iy$$

Define:

$$\mathrm{Re}(z) := x, \quad \mathrm{Im}(z) := y$$

These extract the two coordinates of the number.

Conjugates

Define:

$$\bar{z} = x - iy$$

Then:

$$z\bar{z} = (x + iy)(x - iy) = x^2 + y^2 \in \mathbb{R}$$

This converts a complex number into a real quantity.

Numbers as Points

The expression $x + iy$ behaves like a point (x, y).

Thus $\mathbb{C}$ can be viewed as a plane.

Trigonometry and Direction

On the unit circle:

$$x^2 + y^2 = 1$$

define:

$$(\cos\theta, \sin\theta)$$

Thus:

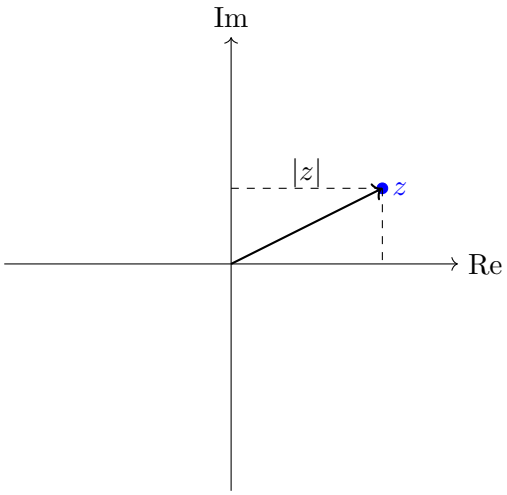
$$\cos^2\theta + \sin^2\theta = 1$$

Each angle θ represents a direction.

Theory

Deeper into $\mathbb{C}$

1. **Complex Plane**
   Each $z = x + iy$ corresponds to a point (x, y).
2. **Modulus**
   $$|z| = \sqrt{x^2 + y^2}$$
   This is the distance from the origin.
3. **Argument**
   If $z \neq 0$, there exists θ such that:
   $$z = |z|(\cos\theta + i\sin\theta)$$
   Each number has magnitude and direction.
4. **Multiplication**
   Let:
   $$z_1 = r_1(\cos\theta_1 + i\sin\theta_1), \quad z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$$
   Then:
   $$z_1 z_2 = r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$$
   Multiplication scales lengths and adds angles.
5. **No Order Relation**
   There is no consistent way to define $<$ or $>$ on $\mathbb{C}$.
6. **Algebraic Closure**
   Every polynomial equation has a solution in $\mathbb{C}$.



Problems

Thinking in $\mathbb{C}$

1. Let $z = a + bi$. Show that:
$$z = 0 \iff a = 0 \text{ and } b = 0$$
2. Show that complex addition corresponds to vector addition in the plane.
3. Show that multiplication by i rotates any complex number by $90°$.
4. Describe geometrically the set:
$$\{z \in \mathbb{C} : |z| = r\}$$
5. Describe:
$$\{z \in \mathbb{C} : |z - a| = r\}$$
What does this represent?
6. Show that:
$$|z + w| \leq |z| + |w|$$
Compare this with the real case.
7. Show that:
$$|z - w| = |w - z|$$
What geometric meaning does this have?
8. Find all $z \in \mathbb{C}$ such that:
$$|z| = |z - 1|$$
(What geometric object is this?)
9. Show that if $|z| = 1$, then:
$$\bar{z} = \frac{1}{z}$$
10. Show that:
$$z + \frac{1}{z} \in \mathbb{R} \quad \text{if } |z| = 1$$

11. Solve:
$$z^2 = -1$$
Then solve:
$$z^2 = i$$
What pattern do you observe?
12. Show that every complex number has exactly two square roots (except 0).
13. Show that:
$$|z|^2 = z\bar{z}$$
Why is this important?
14. Show that multiplication in $\mathbb{C}$ corresponds to:
    - scaling (changing magnitude)
    - rotation (changing angle)
15. Challenge: Find all $z \in \mathbb{C}$ such that:
$$z^3 = 1$$
Describe their geometric arrangement.
16. Challenge (very important): Show that there is no ordering on $\mathbb{C}$ compatible with addition and multiplication.
Why does this mean $\mathbb{C}$ is fundamentally different from $\mathbb{R}$?
17. Challenge (insight): Consider repeated multiplication by i:
$$1, i, -1, -i, 1, \ldots$$
Explain this cycle geometrically.

Remarks

Complex numbers unify algebra and geometry:

- Addition behaves like vector addition
- Multiplication corresponds to rotation and scaling
- i represents a 90° rotation

Remarks

Comparison of Number Systems

Property	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	$\mathbb{C}$
Addition	✓	✓	✓	✓	✓
Subtraction	×	✓	✓	✓	✓
Division	×	×	✓	✓	✓
Density	×	×	✓	✓	✓
Completeness	×	×	×	✓	✓
Algebraic Closure	×	×	×	×	✓

Problems

Thinking in $\mathbb{C}$ (Core)

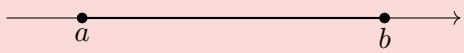
1. Solve $x^2 + 1 = 0$ in $\mathbb{C}$
2. Show $|z|^2 = z\bar{z}$
3. Show multiplication by i rotates by $90°$
4. Solve $z^2 = i$
5. Find all z such that $|z| = 1$

Problems

Parting Shots

1. Show $\sqrt{2}$ is irrational
2. Find a rational between $\frac{1}{3}$ and $\frac{1}{2}$
3. Show $\mathbb{Q}$ is dense
4. Show $\mathbb{Q}$ is countable
5. Show $\mathbb{R}$ is uncountable
6. Construct an irrational number
7. Show $\mathbb{R}$ has no gaps
8. Solve $x^2 = -1$
9. Solve $z^2 = -4$
10. Solve $z^3 = 1$
11. Show $|z|^2 = z\bar{z}$
12. Show $|zw| = |z||w|$
13. Describe $|z| = 1$
14. Describe $|z - 1| = 2$
15. Show $\mathbb{C}$ has no order relation
16. Find all solutions of $z^2 = i$
17. Show every quadratic has solutions in $\mathbb{C}$
18. Challenge: classify $0.1010010001\ldots$
19. Challenge: construct infinitely many rationals between two numbers
20. Challenge (diagram):

All numbers between a and b



Homework aside from Problems

Write a 300-400 word 3 para essay:

- How it was pedagogically and mathematically to do this. - What can be improved - What all in Math/CS Theory y'all would like next

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A Note from the Author

This handout was created independently.

For questions, solutions or improvements:

- Email: `cowboyhehe1@gmail.com`
- GitHub: `www.github.com/Lostless1907`

Please send:

- Clear images or typed work
- Suggestions or corrections

Read the `README.md` before using the source.

This is a living document, feedback and change is welcome.

"And you, my father, there on the sad height,
Curse, bless, me now with your fierce tears, I pray.
Do not go gentle into that good night.
Rage, rage against the dying of the light."
— ***With Love, VPS***